

# Sprout AI

AI-powered market & transport  
intelligence for smallholder farmers.

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# OUR TEAM

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# THE PROBLEM

Smallholder farmers lose 20–40% of potential income because:



They sell to the nearest buyer, missing better-paying markets.



Transport is costly, unreliable, and uncoordinated, limiting market reach.



They cannot assess whether to store or sell for higher returns.



Existing tools give generic price lists, not geospatial or actionable insights.



# SOLUTION & HOW IT WORKS



## Market Intelligence

AI-powered SMS insights deliver real-time market prices, demand forecasts, and store-vs-sell recommendations using LSTM forecasting and geospatial analysis



## Sprout Transport

Affordable logistics through route optimization, order bundling & clustering, eliminating empty return trips and reducing transport costs by up to 40%

### FARMER JOURNEY

01

## Farmer Onboarding

Access via USSD, Web, or Mobile App

02

## AI-Driven Insights

Receive SMS with:

- Best market
- Price advantage
- Store-vs-sell guidance
- Transport quote

03

## Transport Booking

Book optimized transport with bundling & clustering

04

## Delivery & Income

Produce delivered to profitable market, maximizing farmer income

# DATA & MODEL

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## Data Sources

- FAOSTAT
- WFP
- World Bank
- Kenya Met Dept
- Google Earth Engine
- Google Maps Distance Matrix API



## Model Approach

Sprout AI leverages historical and real-time data to deliver actionable insights:

- LSTM price forecasting
- Geospatial yield + surplus modeling
- Market recommendation engine
- Store-vs-sell model
- Route optimization + scheduling

# FEASIBILITY

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**Market Price Differences**

**40–80%**

Price variation across nearby markets



**Transport Costs**

**Up to 35%**

Of farmer marketing expenses



**Informal Transport**

**70%+**

Rely on informal, unpredictable transport



**Poor Timing Losses**

**15–30%**

Post-harvest losses from poor timing



# ADOPTION LOGIC



## Who Would Adopt

Target Users



Smallholder Farmers



Cooperatives & Groups



Aggregators & Processors



Market Buyers



## Why They Adopt

Key Benefits



Access to higher-paying markets



Reduced transport costs



Clear store-vs-sell guidance



Increased reliability



Better market forecasts



## Current Barriers

Challenges



**Fragmented Networks**

Disconnected logistics



**Limited Visibility**

No price transparency



**High Costs**

Expensive coordination



**Lack of Tools**

No simple solutions



# STAKEHOLDERS & DEMO FLOW

## Key Stakeholders



**Farmers**  
(end users)



**Transport Partners**  
or in-house transport team



**Cooperatives**  
and farmer groups



**Market Buyers**  
/ off-takers

## Demo Flow

1

### Input

Farmer selects crop and location via USSD. System retrieves market prices, forecasts, and distance data



2

### Model Processing

LSTM model predicts price trends. Surplus/deficit model identifies best markets. Store-vs-sell engine evaluates timing



3

### Output

Recommended market, expected price advantage, store-vs-sell guidance, transport quote and pickup option



4

### Meaning

Farmer knows **where** to sell, **when** to sell, and **how** to get produce there affordably



# ACHIEVEMENTS AND NEXT STEPS



## Achievements

- ✓ Desktop Research  
Data & Publications
- ✓ User Engagement  
Active participation
- ✓ Farmer Interface  
User-friendly platform
- ✓ Output Generation  
Insights, Route Optimization, Clustering & Bundling,  
Cost Computation
- ✓ Voice Integration  
Voice-enabled features



## Next Steps

1

### Market Research

Comprehensive market analysis



2

### Pilot Program

1,000 farmers across 5 counties



3

### Partnerships

Stakeholders, Innovators, Donors

# JOIN US

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Sprout AI provides market intelligence and transport optimization for smallholder farmers.

We invite partners to join us in scaling Sprout AI.

**Let's grow together**

[contact@sproutai.co.ke](mailto:contact@sproutai.co.ke)